

3 R intro; Epidemic Data and Time Series Tools



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 3

R intro; Epidemic Data and Time Series Tools
Monday 22 September 2025

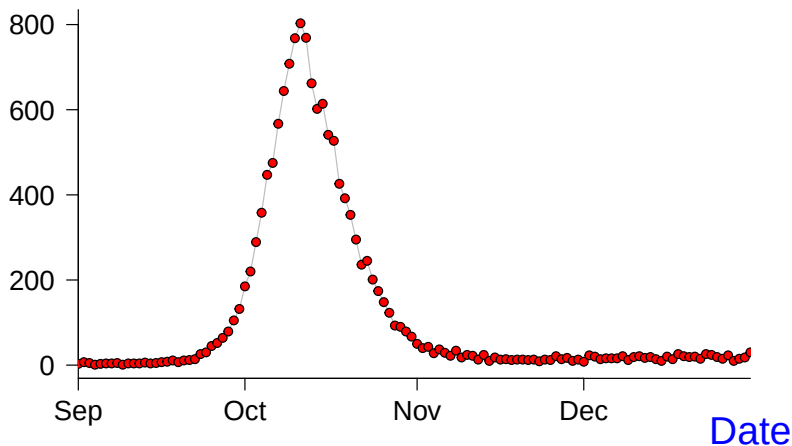
Announcements

■ **Assignments:**

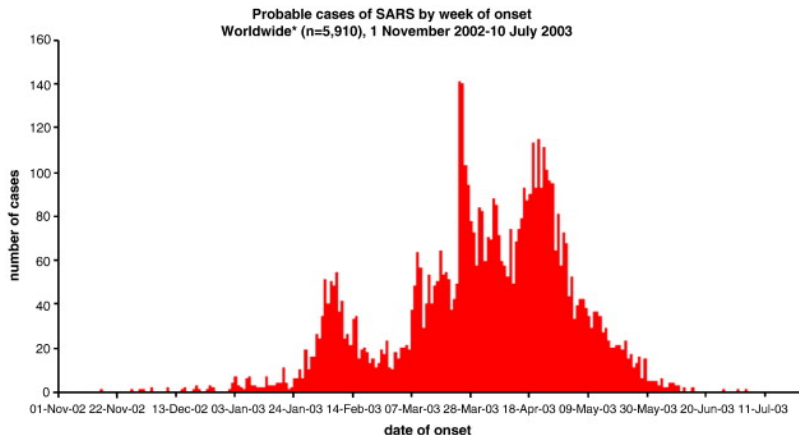
- Assignment 1 due 24 Sep 2025
- Assignment 2 due 8 Oct 2025
(good to work on before class on 29 Sep 2025)

P&I Mortality, Philadelphia, 1918

P&I Deaths

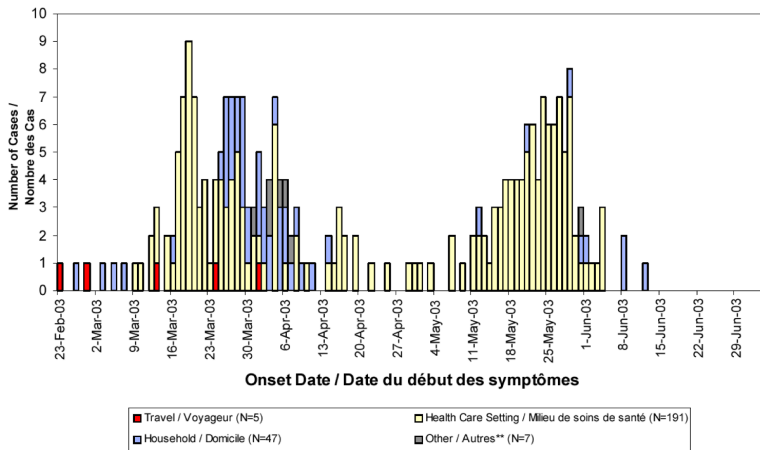


SARS-CoV-1 in 2003 (Worldwide)



*This graph does not include 2,527 probable cases of SARS (2,521 from Beijing, China), for whom no dates of onset are currently available.

SARS-CoV-1 in 2003 (Toronto)



$N = 249$ (of 250 reported)

Some SARS-CoV-1 Facts

- High case fatality
 - 1918 flu $< 3\%$
 - SARS-CoV-1 $> 10\%$
- Long hospital stays
 - Mean time from admission to discharge or death:
~ 25 days in Hong Kong
- 8098 probable cases, 774 deaths
- How bad would it have been if it had not been controlled?

COVID (ancestral) hospitalization and survival in Ontario

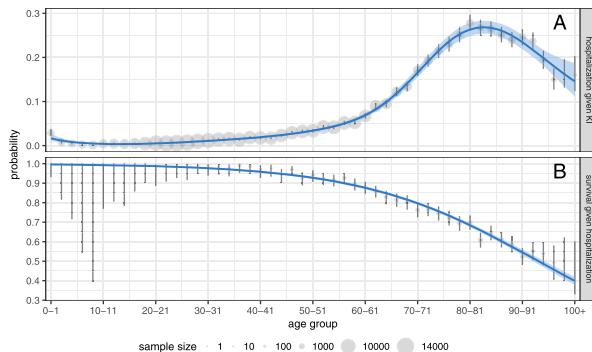
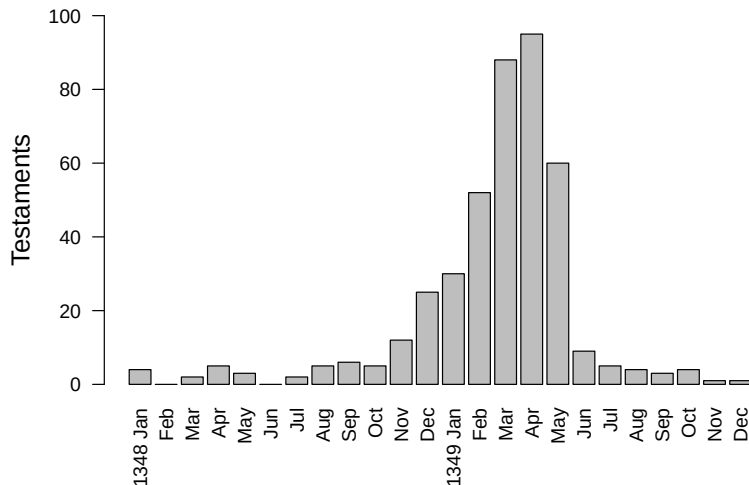


Fig. 4 Age-dependent COVID-19 hospitalization probability for known SARS-CoV-2 infection (panel **a**) and survival probability for hospitalized patients (panel **b**) in Ontario. We give age-by-age estimates of each probability (points; 95% exact binomial confidence intervals given by vertical lines), where point area is proportional to age-specific sample size. We additionally provide more precise estimates of these probabilities under stricter assumptions, modelling the hospitalization probability using a generalized additive model and the survival probability using a generalized linear model (curves; 95% confidence bands given by shaded regions). See "Methods" section for details

Papst et al (2021), BMC Public Health, 21:706

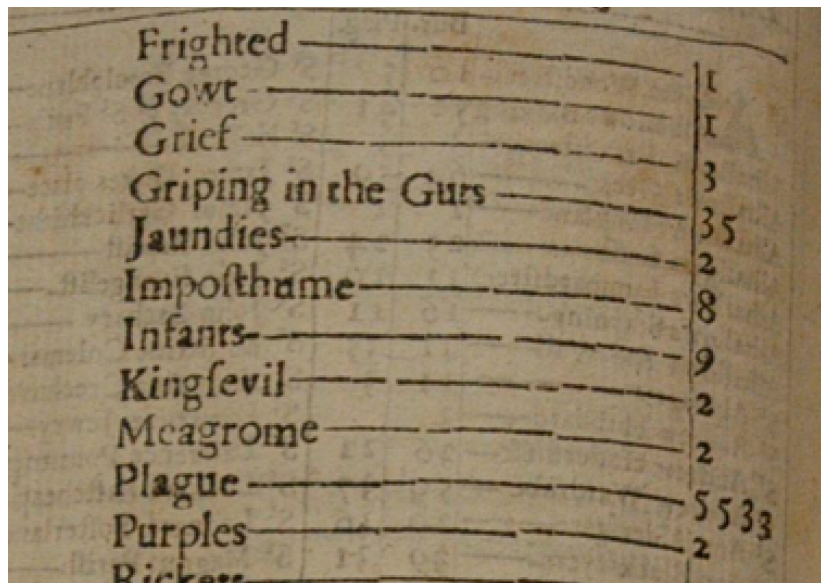
The Black Death in London, England, 1348–1349



London Bill of Mortality, 26 Sept to 3 Oct 1665

The Diseases and Casualties this Week.		London 41. From the 26 of September to the 3 of October. 1665	
	Frighthead	1	1
	Goyle	1	1
A	Grief	1	1
	Gripping in the Guts	35	35
B	Jaundies	2	2
	Impoethame	2	2
C	Infants	2	2
	Kingsevil	2	2
D	Meargrome	2	2
	Plague	553	553
E	Purples	2	2
	Rickets	2	2
F	Rising of the Lights	2	2
	Rupture	13	13
G	Scurvy	1	1
	Spotted Fever	65	65
H	Stilborn	10	10
	Stone	1	1
I	Stopping of the Stomach	6	6
	Suddenly	1	1
J	Surfeit	36	36
	Teeth	11	11
K	Thrush	3	3
	Tifick	5	5
L	Vomiting	4	4
	Winde	1	1
M	Wormes	12	12
Decreased in the Burials this Week		1837	
Parishes clear of the Plague		7	
Parishes Infected		123	
The Affize of Bread (as forth by Order of the Lord Mayor and Court of Aldermen, A penny Wheaten Loaf to contain Nine Ounces and a half, and three half-penny White Loaves the like weight.			

London Bill of Mortality, 26 Sept to 3 Oct 1665



Frighted	1
Gowt	1
Grief	3
Griping in the Gurs	35
Jaundies	2
Imposthume	8
Infants	9
Kingsevil	2
Meagrome	2
Plague	5533
Purples	2
Rickets	

Mortality Bills are typically handwritten

LONDON 29. From the 4. of July to the 11. of August 1665.											
Buried, Plag.			Buried, Plag.			Buried, Plag.			Buried, Plag.		
St Alban Woodstreet	2	1	St Clement Eastcheap	1		St Margaret Newfish			St Michael Crookedla	4	3
Alhallowes Bark-			St Dionis Backchurch	2		St Margaret Patrons			St Michael Queenhit	2	
Alhallowes Breadstreet	1		St Dunstons East			St Mary Abchurch	1		St Michael Quern		
Alhallowes Great			St Edmund Lombardst			St Mary Aldermanbury			St Michael Royal		
Alhallowes Honilane	1		St Ethelborough	2		St Mary Aldemary			St Michael Woodstreet		
Alhallowes Lefi			St Faiths	1		St Mary le Bow			St Mildred Breadstreet		
Alhallowes Lombardst			St Gabriel Fenchurch			St Mary Bothaw			St Mildred Poultry		
Alhallowes Staining	1		St George Botolphane			St Mary Colechurch			St Nicholas Acon		
Alhallowes the Wall	4	3	St Gregories by St Paul			St Mary Hill			St Nicholas Coleabby		
St Alphage			St Hellen	2	1	St Mary Mag. Milkstr			St Nicholas Olaves		
St Andrew Hubbard			St James Dukes place	1		St Mary Mag. Oldfish			St Olave Hartstreet		
St Andrew Underhafe	3		St James Garlickhithe	1		St Mary Mounthaw			St Olave Jewry		
St Andrew Wardrobe	1		St John Baptst			St Mary Summerfet	2	1	St Olave Silverstreet	4	1
St Anne Aldergate	1		St John Evangelist			St Mary Stainings			St Pancras Soperlane		
St Anne Blackfryers	7	6	St John Zichary			St Mary Woolchurch			St Peter Cheap		
St Antholins Parish	1		St Katharine Coleman	1		St Mary Woolnoth			St Peter Cornhil		
St Austins Parish			St Katharine Creechur			St Martins Iremongerl			St Peter Paulwharf		
St Barthol. Exchange	1		St Lawrence Jewry			St Martins Ludgate	2	1	St Peter Poor	1	
St Bennet Fynck			St Lawrence Pountney			St Martins Orgars			St Steven Colemanstr	2	1
St Bennet Gracechurch	2		St Leonard Eastcheap			St Martins Outwich	1		St Steven Walbrook		
St Bennet Paulwharf	7		St Leonard Fosterlane			St Martins Vintrey			St Swithin	2	1
St Bennet Sherehog			St Magnus Parish	1		St Matthew Frydaystr			St Thomas Apostle	1	1
St Boroloph Billingsgate			St Margaret Lothbury			St Michael Balishaw	5	4	Trinity Parish	1	
Christ Church	5	3	St Margaret Moses			St Michael Cornhil			St Vedast alias Fosters		
St Christophers											
Christened in the 9 Parishes within the walls			Buried			86			Plague		
St Andrew Holborn	00	40	St Boroloph Aldergate	11	9	St George Southwark	13	4	St Sepulchres Parish	117	81
St Bartholomew Great	4	4	St Boroloph Aldgate	24	4	St Giles Cripplegate	105	42	St Thomas Southwark	7	5
St Bartholomew Lefi			St Boroloph Bishopsgate	37	29	St Olave Southwark	20	6	Trinity Minories		
St Bridget	24	14	St Dunstan West	19	9	St Saviour Southwark	21	1	At the Pesthouse	6	6
Bridewel Precinct	1	1									
Christened in the 16 Parishes without the walls			Buried			473			Plague		
Christs Church			St Kath. near the Tower	5	1	St Mary Islington	3	2	St Paul Shadwel		
St John at Hackney	1		Lambeth Parish	21	13	St Mary Newington	16	5	Rotherhithe Parish	7	3
St Giles in the Fields	208	215	St Leonar d Shoreditch			St Mary Whitechappel			Stepney Parish	47	1
St James Clerkenwel	5		St Magdalen Bermond.	14							
Christened in the 12 Out-parishes in Middlesex and Surrey			Buried			455			Plague		

But handwriting is usually very clear

A historical document from the Great Plague of London, 1665, showing a table of burials and deaths by parish. The document is titled "LONNDRN 29th" and lists the following parishes and their corresponding numbers of burials and deaths:

	Buried.	Plag.
St Alban Woodstreet	2	1
Alhallows Bark.	2	
Alhallows Breadstreet	1	
Alhallows Great		

But handwriting is usually very clear

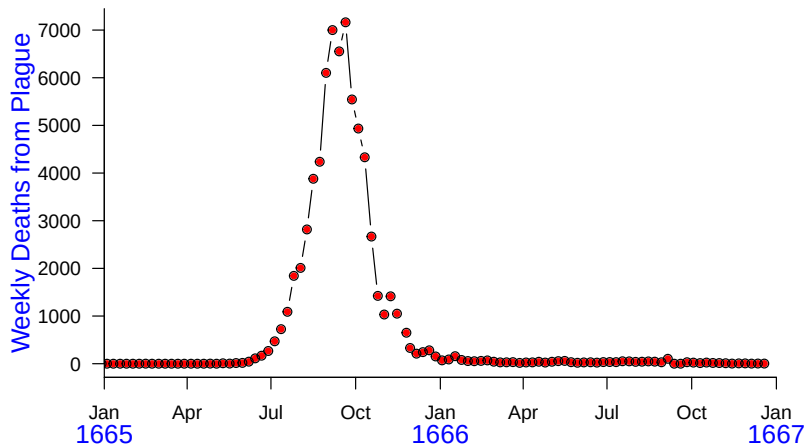
St Christophers

Christned in 97 the Parishes

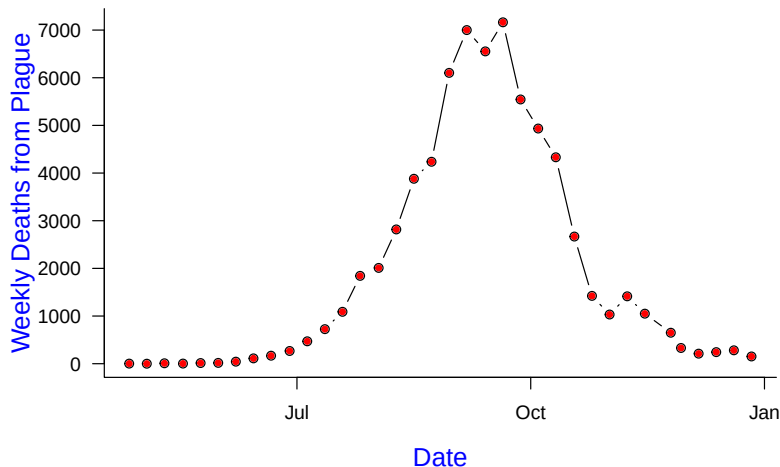
St Andrew Holborn—	66	40	St
St Bartholomew Great	7	4	St
St Bartholomew Less—			St
St Bridget—	24	17	St
Bridewel Precinct—	1	1	

Christned in the 16 Parishes

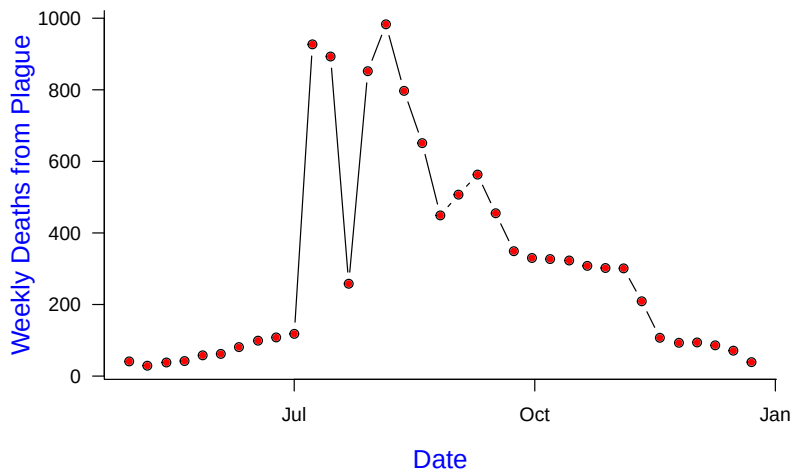
The Great Plague of London, 1665



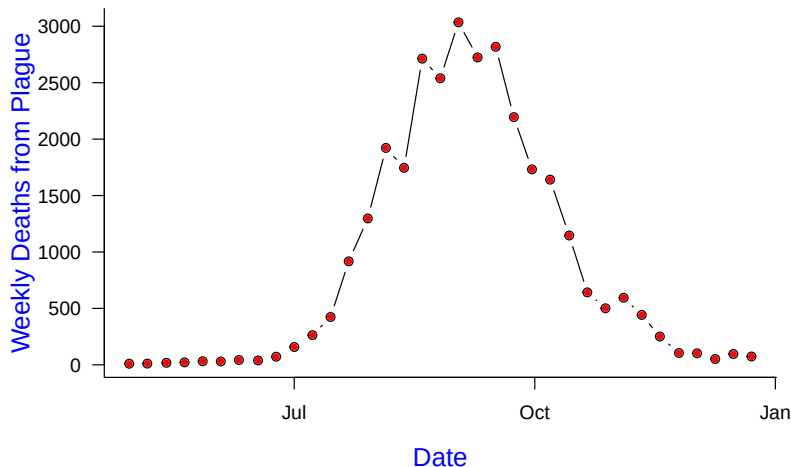
The Great Plague of London, 1665



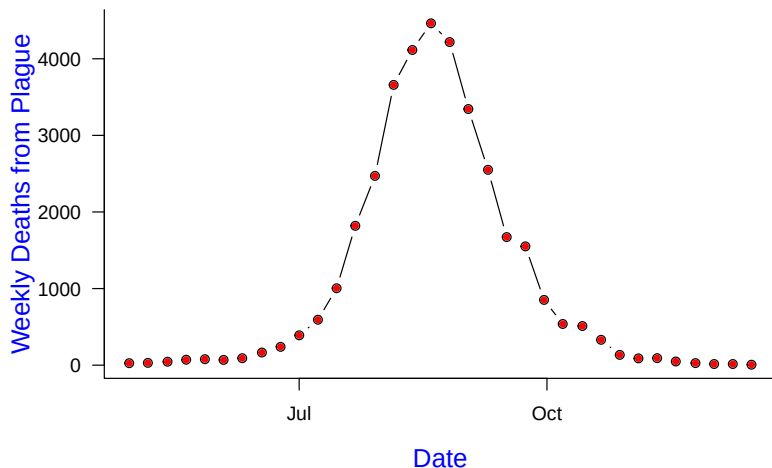
London Plague of 1593



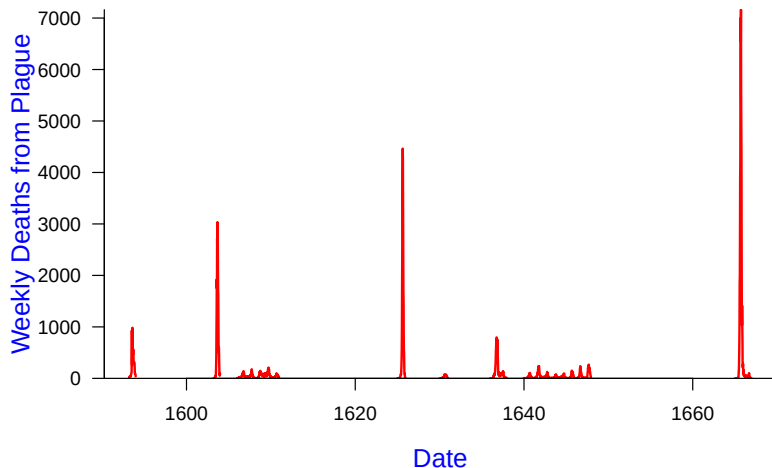
London Plague of 1603



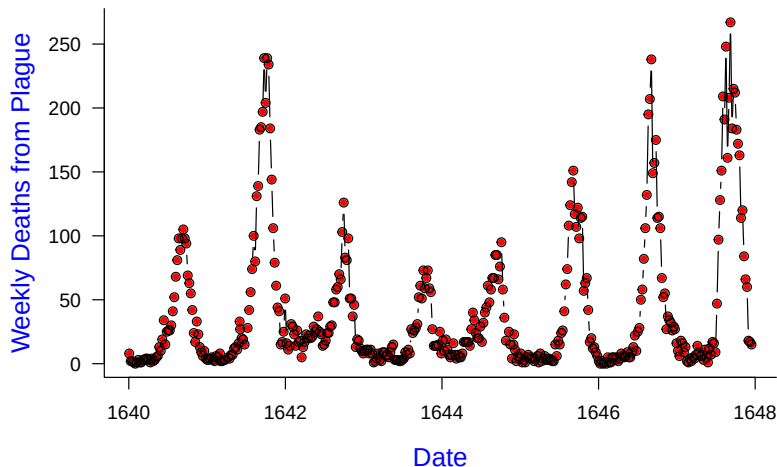
London Plague of 1625



Weekly Deaths from Plague in London, 1592–1666



Weekly Plague in London, 1640–1648



Some Plague Facts

- Plague epidemics recorded from Roman times to early 1900s.
- $\gtrsim 1/3$ Europe's population died in "Black Death" of 1348
 - ~ 300 years for the population to reach the same level.
- Recently (2011) established (at McMaster!) that the pathogen that caused The Black Death was *Yersinia pestis*

[Bos *et al.* 2011, *Nature* **478**, 506–510]

- More recently (2014) established (again at McMaster!) that the pathogen that caused The Plague of Justinian (541–543 AD) was *Yersinia pestis*

[Wagner *et al.* 2014, *Lancet Infectious Diseases* **14**, 319–326]

- *Y. pestis* still a concern?
Yes: Rodent reservoir, antibiotic-resistant strains, bioterrorism
- **Spatial data** for any plagues? Yes, for London in 1665...

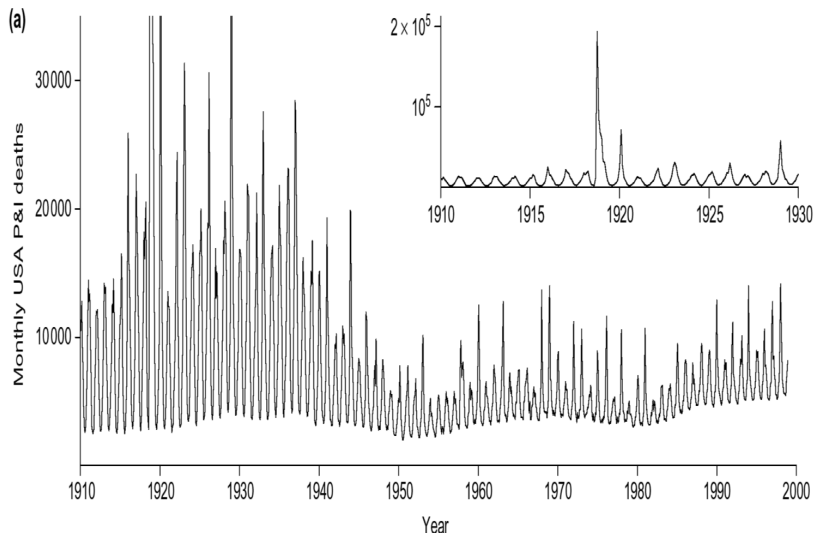
Visualization of spatial structure of Great Plague

- GIS encoding of parish boundaries
- Overlay parish boundaries on more modern map for reference
- Colour parishes as they become infected
- Is there evidence for spatial spread or was the spatial pattern random?
- DE low-tech animation...
- CBC high-tech animation...
 - *The Nature of Things*, 21 August 2014.
<https://www.imdb.com/title/tt8746582/>

Visualization of entire course of the Great Plague

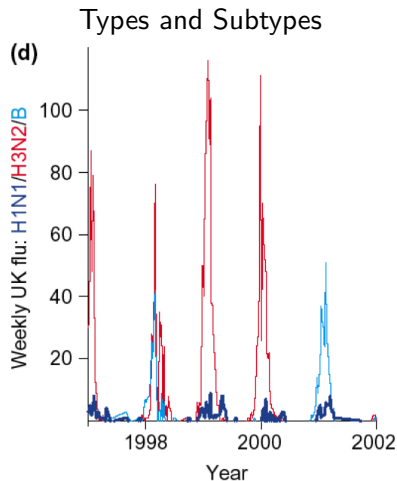
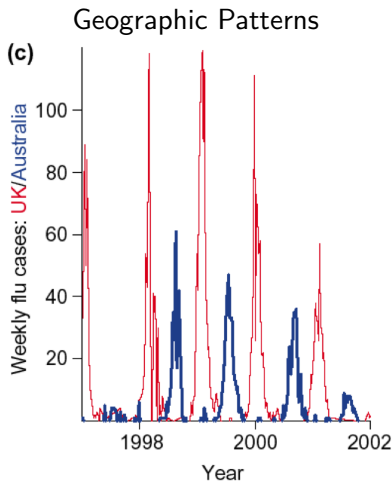
- What happened after initial spatial spread?
- Visualize full spatial epidemic structure
- Show magnitude of epidemic in each parish with cylinder.
- Epidemic Visualization (EpiVis) software by Junling Ma.
- Full spatial dynamics of the Great Plague (7 fps, 40 fps)
 - there is one frame per day, so 7 fps means *one week per second*

P&I mortality in U.S.A., 1910–1998



Earn, Dushoff & Levin 2002, *Trends in Ecology and Evolution* **17**, 334–340

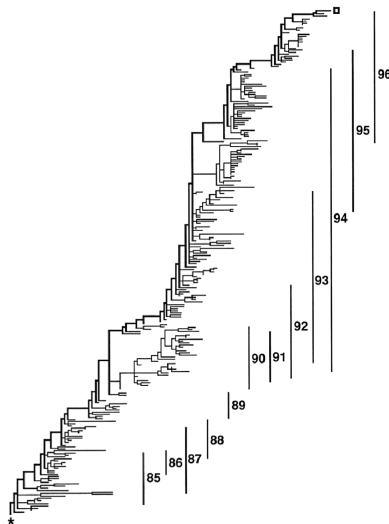
Influenza Incidence Patterns (lab confirmed)



Earn, Dushoff & Levin 2002, *Trends in Ecology and Evolution* **17**, 334–340

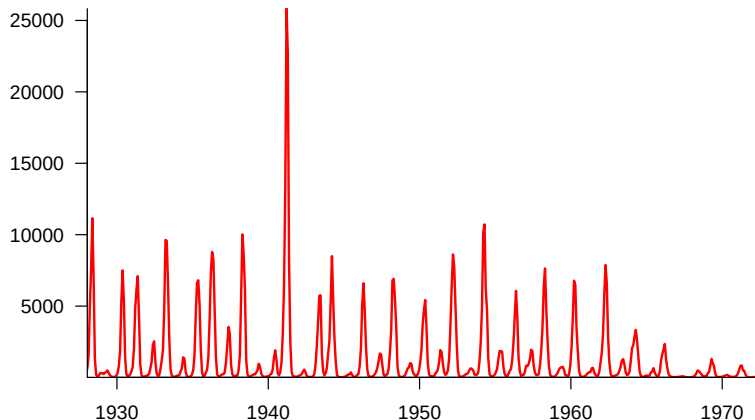
Influenza Evolution

Molecular
phylogenetic
reconstruction of
influenza A/H3N2
evolution,
1985–1996
(Fitch *et al.* 1997)



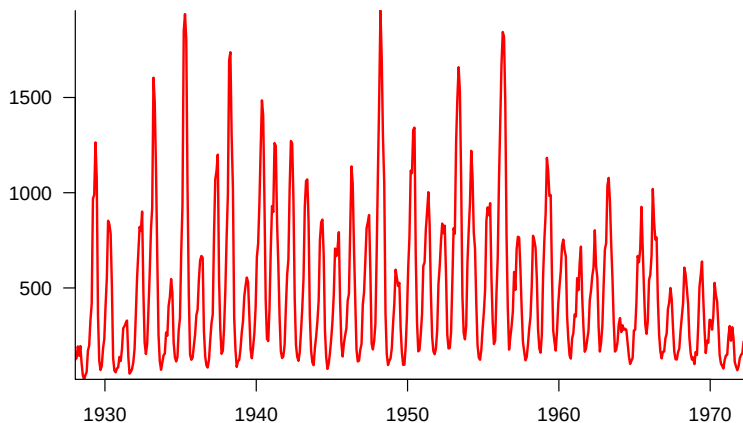
Measles in New York City, 1928–1972

Monthly Cases



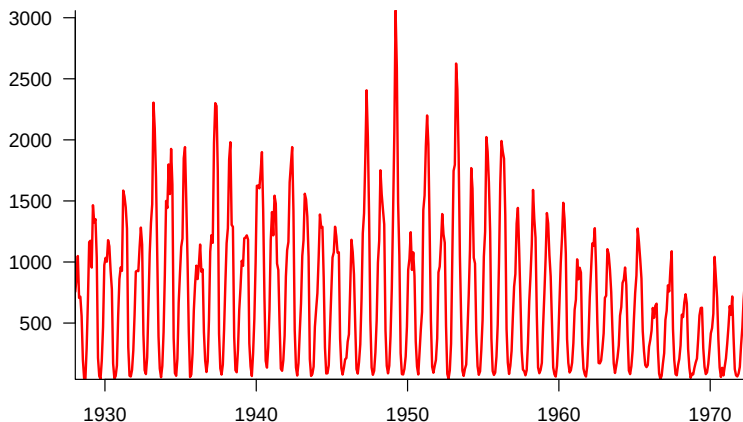
Mumps in New York City, 1928–1972

Monthly Cases

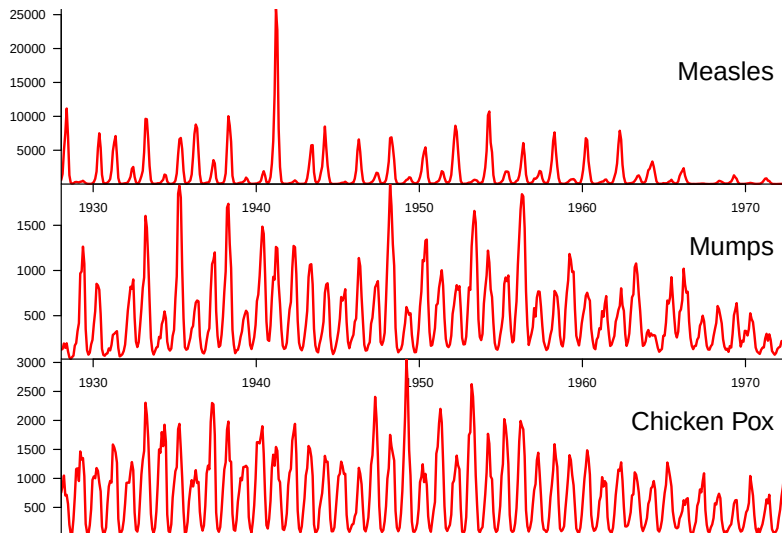


Chicken Pox in New York City, 1928–1972

Monthly Cases

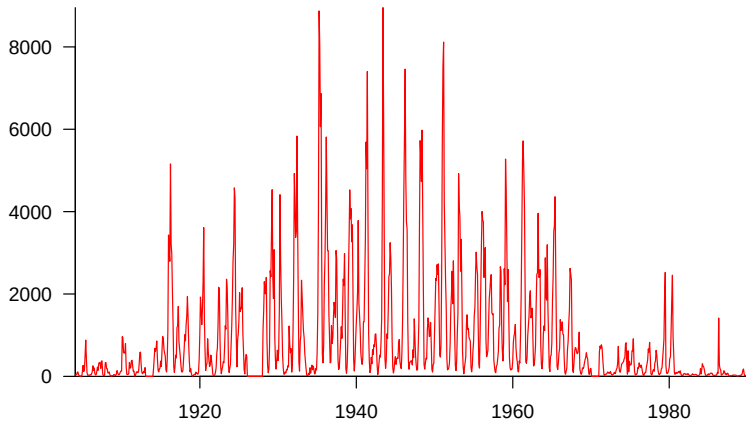


Childhood diseases in New York City, 1928–1972



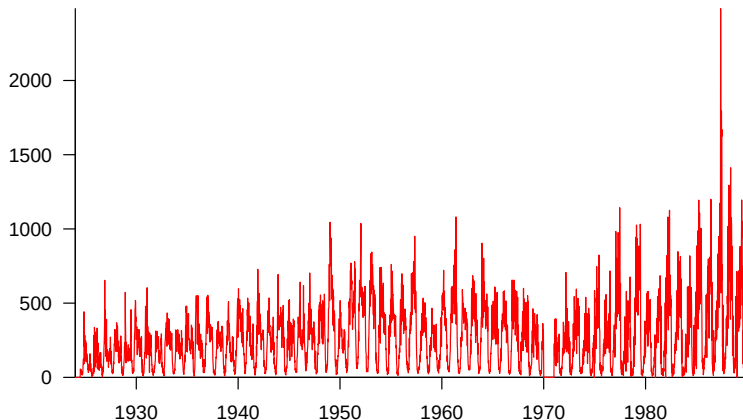
Measles in Ontario, 1904–1989

Monthly Cases



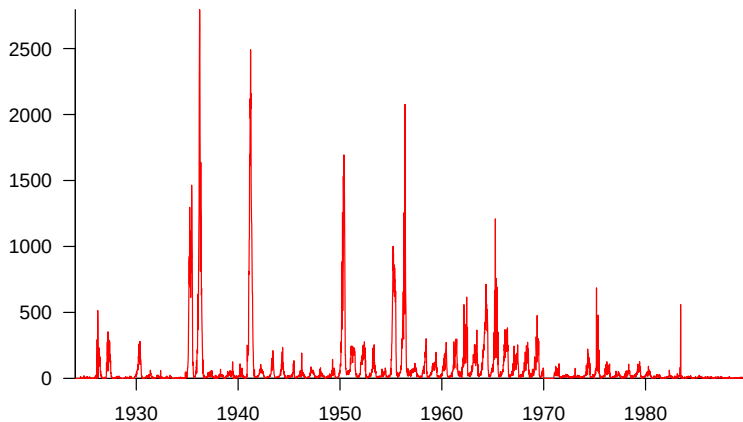
Chicken Pox in Ontario, 1924–1989

Monthly Cases



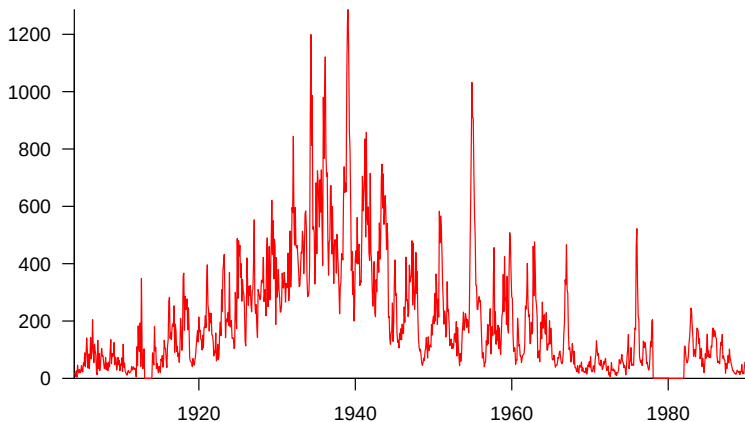
Rubella in Ontario, 1924–1989

Weekly Cases

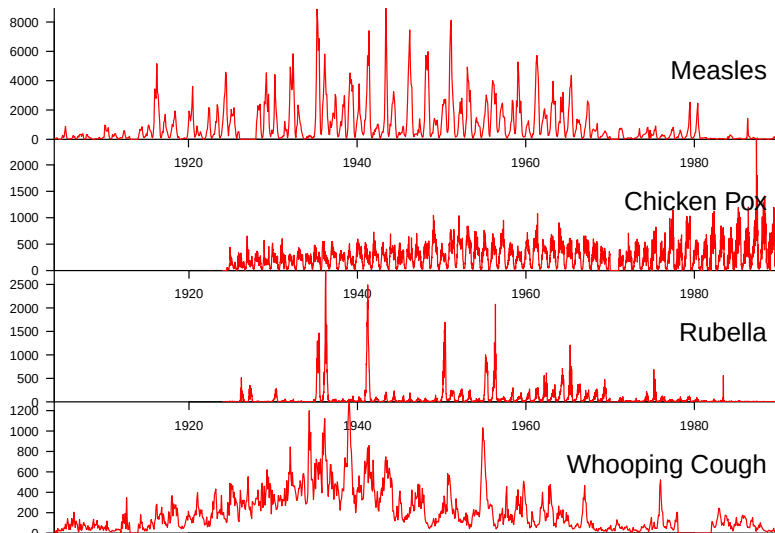


Whooping Cough in Ontario, 1904–1989

Monthly Cases



Childhood diseases in Ontario, 1904–1989



Ontario Disease Notification Data

Province of

YEAR: 1939

COUNTY.....

MUNICIPALITY.....

Month	Week End.	CSM		C.P.		DIP.		DYS. A/B		EN. LETH.		ERY.S.		G.C.		FLU.		INF. JAUN.		G.M.		MEAS.		MUMPS		PAR. TYPH.		
		C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	
Jan.	7 1			452	1	3	0	1	0			5	1	101	0	8	1	17	0	17	0	670	1	56	0	2	0	
	14 2	2	1	490	0	8	0					5	0	82	0	21	1	18	0	18	0	850	0	92	0	1	0	
	21 3	2	1	511	0	9	3			0	1	5	0	89	0	16	2	26	0	22	0	932	0	98	0			
	28 4	1	0	384	0	2	0					2	0	73	0	164	0	10	0	28	0	933	1	24	0			
	Total	5	2	1837	1	24	3	1	0	0	1	17	1	352	0	263	4	71	0	85	0	3385	2	270	0	3	0	
Feb.	4 5			355	0	7	1	1	0			3	0	83	0	57	1	24	0	25	0	1335	1	110	0	2	0	
	11 6	2	1	363	0	1	0	1	0			7	0	82	0	27	1	41	1	29	0	1033	0	91	0	1	0	
	18 7	2	1	354	1	2	0					4	1	68	0	103	1	35	0	44	0	1161	0	59	0			
	25 8	1	1	308	0	2	0					9	0	56	0	177	0	19	0	28	0	999	0	73	0			
	Total	5	3	1380	1	12	1	2	0			23	1	289	0	364	3	119	1	126	0	4588	1	333	0	3	0	
Mar.	4 9	1	1	271	0	7	1	3	1			7	0	93	0	114	19	21	0	40	0	1131	2	169	0	1	0	
	11 10			239	0	7	0	2	0			8	1	61	0	137	8	31	0	32	0	845	0	91	0	2	0	
	18 11			166	0							6	0	66	0	132	6	5	0	59	0	969	2	69	0	1	0	
	25 12	1	2	236	0	1	0	1	0			7	0	63	0	306	16	9	0	20	0	879	0	170	0	1	0	
	Total	2	3	912	0	15	1	6	1			28	1	283	0	463	49	66	0	151	0	3884	4	389	0	34	0	
Apr.	1 13	2	0	139	0	3	0	1	0			8	0	93	0	667	6	1	0	24	0	950	0	89	0	3	0	
	8 14	2	0	162	0	1	0	1	0			5	0	67	0	731	22			14	0	770	0	65	0	1	0	
	15 15	2	0	108	0	1	0			0	1	11	0	41	0	529	16	2	0	16	0	745	0	56	0			
	22 16	1	1	134	0	2	0	1	0		1	1	6	0	64	0	245	8	2	0	26	0	845	0	54	0		
	29 17	5	1	167	0	4	0	2	0	2	1	3	0	55	0	124	9	2	1	13	0	746	1	120	0			
Total	12	2	710	0	11	0	5	0	3	3	33	0	322	0	2114	61	9	1	99	0	4016	1	284	0	4	0		
6 18	2	0	104	0	1	0	2	0			4	0	71	0	76	3	1	0	14	0	877	0	63	0	3	0		

Dominion Bureau of Statistics Disease Notification Data

VITAL STATISTICS BRANCH - COMMUNICABLE DISEASE SECTION

Cases of *H. influenzae* Reported by Provincial Health Departments, Year *1924*

WEEK ENDING	P.E.I.		N.S.		N.B.		QUE.		ONT.		MAN.		SASK.		ALTA.		B.C.		CANADA	
	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT	WIS	WIT
<i>Jan 5</i>			11										1						12	
2 12			29										18						47	
3 19			37										32						69	
4 26			75	152			68	181	36	13	64		97		4	88	602			
<i>5 FEB 2</i>			12		1								53						66	
6 9			5										40						45	
7 16			31										14						45	
8 23			2	50	1	2	267	202	48	4	111		116		1	7	797			
<i>9 MAR 1</i>			2										21						23	
10 8													9						9	
11 15			3										11						14	
12 22			60										34						94	
13 29			2	67			144	140	52	15	90		15		7	17	515			
<i>14 APR 5</i>			9										11						20	
15 12			1										12						13	
16 19			26		1								8						35	
17 26			14	50	3	4	42	140	39	16	47		67		5	33	394			
<i>18 MAY 3</i>			26										2						28	

All Historical Canadian Infectious Disease Data

<https://canmod.net/digitization/>

Over a Century of Infectious Disease Surveillance in Canada

<https://www.medrxiv.org/content/10.1101/2024.12.20.24319425v1>